PROBLEM STATEMENT 02

Build a Benchmarking Platform for AI Agents, Not Just Models

Traditional AI benchmarks measure model performance, but they rarely explain how well an entire AI agent performs. Two agents using the same model can produce very different results because of their prompts, tools, memory, and retry strategies: one may complete a task in a handful of calls, while another consumes significantly more resources and still fails.

Developers need a consistent way to compare agent performance beyond successful demos.

GOAL

Build a framework-independent benchmarking platform that evaluates agents of a selected category (coding, research, or customer support) against identical tasks and measurable performance criteria.

MANDATORY REQUIREMENTS

- **Framework independence:** Support agents built using different frameworks, or no framework, without requiring them to be rewritten.
- **Repeatability:** Execute tasks multiple times to measure reliability rather than depending on a single successful run.
- **Inspection & comparison:** Provide a comparison interface with detailed, step-by-step execution records for individual runs.
- **Consistent evaluation:** Run agents against the same tasks and record success, steps, tool calls, tokens, cost, and execution time.
- **Failure testing:** Introduce controlled failures such as unavailable models, broken tools, or misleading instructions, and evaluate agent behavior.

EXPECTED OUTCOME

A functioning benchmarking platform that allows developers to compare AI agents objectively, understand differences in performance, and inspect how agents behave under normal and failure conditions.

OPTIONAL IMPLEMENTATION GUIDANCE

Consider a leaderboard, repeatable task suites, execution traces, and comparative dashboards. Bifrost can provide gateway-level visibility into model requests and operational metrics; additional instrumentation may be needed to capture agent-specific steps and tool execution.